A Hidden Limit of Space and Time Could Lead to Ultrafast Electronics

From: ScienceDaily Most Popular Mined: 2026-11-09 5 min read

Quantum physics has a famous rule: you cannot know everything about a particle at once. Werner Heisenberg's **uncertainty** principle says that a particle's position and **momentum** cannot both be measured with unlimited precision. That limit is built into the laws of nature, not caused by faulty instruments.

But position and time are different. Heisenberg's rule does not restrict those two in the same way. Now researchers have shown experimentally that another limit exists: an electron's location and its behaviour over time cannot both be pinned down with unlimited precision. This new 'space-time limit' could matter a great deal for quantum technology and ultrafast electronics.

The discovery comes from an international team at the Regensburg Center for Ultrafast Nanoscopy in Germany, led by professors Jascha Repp, Rupert Huber, Franz Giessibl and Klaus Richter, working with a team at the Max Planck Institute in Hamburg led by Angel Rubio.

To watch **electrons** move, the team built a new laser system that works like a high-speed camera. Timed laser pulses drive **electrons** from an extremely sharp metal tip toward a silver surface, just a few atomic diameters away. By changing the gap between two pulses, the researchers can see exactly when the **electrons** move. 'By varying the time interval between the two laser pulses, we can directly observe how the **electrons** respond,' said Simon Maier, the lead author of the paper.

The movements happen over **attoseconds**: one attosecond is one-billionth of a billionth of a second. The difference between an attosecond and a second is comparable to the difference between one second and the entire age of the universe. At these distances, **electrons** move roughly a thousand times faster than atoms and molecules.

What makes the result remarkable is how the **electrons** travel. Instead of moving like tiny balls, they behave as quantum waves, letting them pass through an energy barrier even though classical physics says they should not have enough energy to cross it. This is quantum **tunnelling**, a bit like passing through a solid wall without breaking it. 'Our measurement can be understood as a high-speed camera for the electron wave packets,' said doctoral student and co-author Katharina Glöckl.

Word Treasure

uncertainty -- Not being able to know something exactly or for sure.

momentum -- A measure of how much motion an object has, depending on its mass and speed.

attoseconds -- Extremely short periods of time, each one-billionth of a billionth of a second.

tunnelling -- When a tiny particle passes through a barrier it should not have enough energy to cross.

electrons -- Tiny particles with a negative charge that move around the centre of an atom.

Background you need

Quantum physics is the science of the very smallest things, like electrons, which behave in ways that seem strange compared with everyday objects.

An electron is a tiny particle with a negative charge that moves around the centre of an atom.

A laser is a device that makes a narrow, powerful beam of light, and very short laser pulses can act like the flash of a super-fast camera.

Quantum tunnelling is when a particle passes through an energy barrier that it should not be able to cross, because it behaves like a wave rather than a tiny ball.

Break it down

LEAD: Scientists have measured a new 'space-time limit' for electrons, a step toward ultrafast electronics.

KEY EVENT: An electron's location and its behaviour over time cannot both be pinned down with unlimited precision.

+- WHO: An international team at the Regensburg Center for Ultrafast Nanoscopy, working with the Max Planck Institute in Hamburg.

+- WHEN: The movements happen over attoseconds, one-billionth of a billionth of a second.

EVIDENCE: A new laser system worked like a high-speed camera, showing electrons tunnelling through an energy barrier as quantum waves.

OPEN QUESTION: How will this new limit shape the quantum technology and ultrafast electronics of the future?

Why it matters

Faster electronics could change computers and phones, and this experiment shows that the strange rules of quantum physics set the speed limits for the technology of the future.

Test what you remember

1. What does Heisenberg's uncertainty principle say?

- (A) Light is faster than everything
- (B) Time travel is possible
- (C) A particle's position and momentum cannot both be measured with unlimited precision
- (D) Electrons always move in straight lines

2. What is the new limit about?

- (A) An electron's location and its behaviour over time
- (B) How fast a rocket can fly
- (C) How cold space can get
- (D) How bright a laser can be

3. Where is the research team based?

- (A) London and Paris
- (B) Regensburg, Germany, and Hamburg
- (C) Tokyo and Beijing
- (D) New York and Boston

4. How long is an attosecond?

- (A) One second
- (B) One-millionth of a second
- (C) One-billionth of a billionth of a second
- (D) One-thousandth of a second

5. How do the electrons behave instead of like tiny balls?

- (A) As drops of water
- (B) As tiny magnets
- (C) As beams of light
- (D) As quantum waves

6. What is quantum tunnelling?

- (A) Passing through an energy barrier without enough energy to cross it
- (B) Sending a message through space
- (C) Heating metal until it melts
- (D) Digging a tunnel through a mountain

Answer key (try the quiz first!): 1: C 2: A 3: B 4: C 5: D 6: A

Questions to think about

1. Lead author Simon Maier: He says that by varying the time interval between the two laser pulses, the team can directly observe how the electrons respond.

2. Co-author Katharina Glöckl: She says the measurement can be understood as a high-speed camera for the electron wave packets.

MY THOUGHTS (at least 20 words):